MODULE 3.2

Access Control Concepts

Network Defense | Cisco Networking Academy

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Topics:

- What Zero Trust security actually means and why it exists
- The main access control models — DAC, MAC, RBAC, ABAC, and a couple of others
- Why least privilege matters and how privilege escalation breaks it
- What Network Access Control (NAC) does and why it matters for zero trust

What Is Access Control?

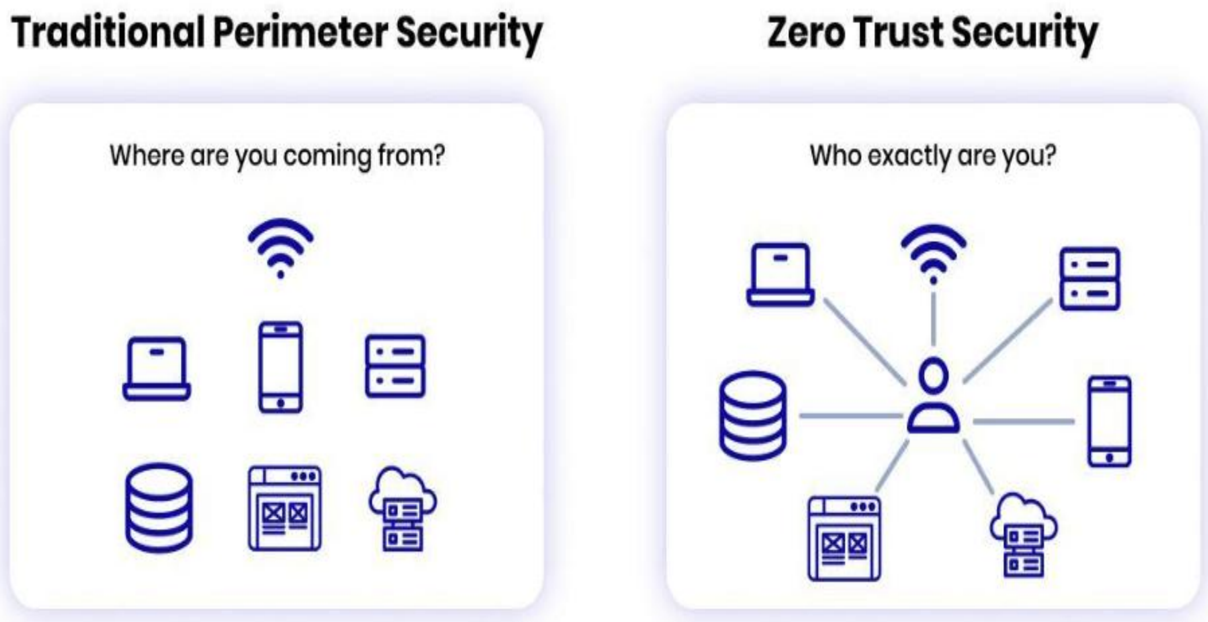
And why "never trust, always verify" matters

So, What Is Access Control?

- Deciding who (or what) is allowed to use a resource
- Applies to people, devices, and apps — anything asking for access
- The goal: keep the wrong people/things out, let the right ones in

The Zero Trust Idea

- Core rule: "Never trust, always verify"
- Old way: trusted if you were already inside the network
- Zero trust way: every request gets checked, every single time
- Logging in once doesn't mean you're trusted for everything else



The Three Pillars of Zero Trust

Workforce, Workloads, and Workplace

Workforce

- People — employees, contractors, anyone using a device to reach work apps
- Zero trust checks that it's the right person on a secure device, no matter where they are

Workloads

- The apps and services running in the cloud or a data center
- Covers things like an API or microservice reaching into a database

Workplace

- Every device connected to the network — laptops, printers, cameras, even smart devices
- Basically anything with a network connection needs to be checked, not just people

Access Control Models

Different ways to decide who gets in

Six models, six different ways to make the same decision:

DAC

- Owner decides who else can access their own data or files

MAC

- Very strict — uses security labels like Top Secret; common in military systems

RBAC

- Access is tied to your job role, not you personally

ABAC

- Looks at multiple attributes at once — user, resource, time, location

Rule-Based

- Access follows fixed rules, like allowed IP addresses or ports

Time-Based

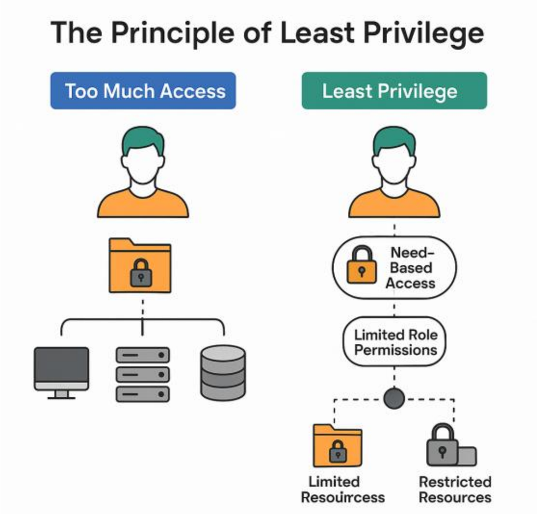
- Access only allowed during set hours or days

Principle of Least Privilege

Give people only what they need — nothing more

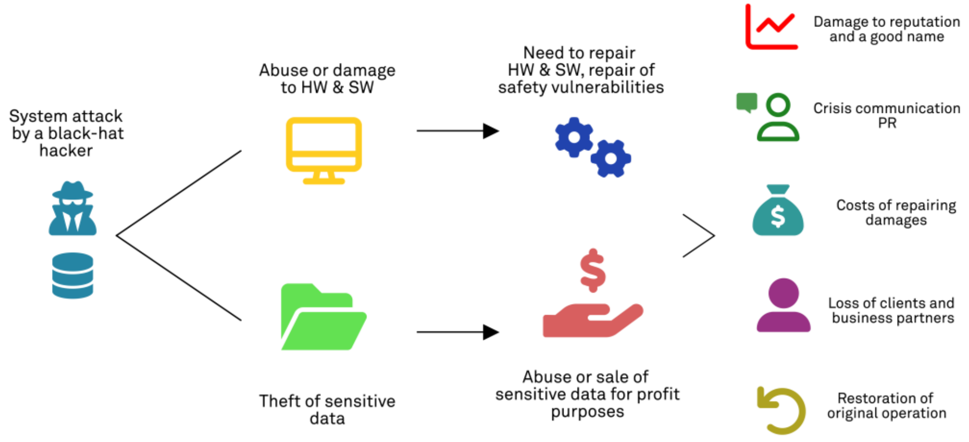
Least Privilege

- Only give users the minimum access needed to do their job
- No extra permissions just sitting around unused
- If an account gets hacked, the damage stays limited to what it could actually touch



Privilege Escalation

- An attack where someone grants themselves more access than they should have
- Usually exploits a bug or weakness in a server or access control system
- This is exactly the kind of attack least privilege is designed to prevent

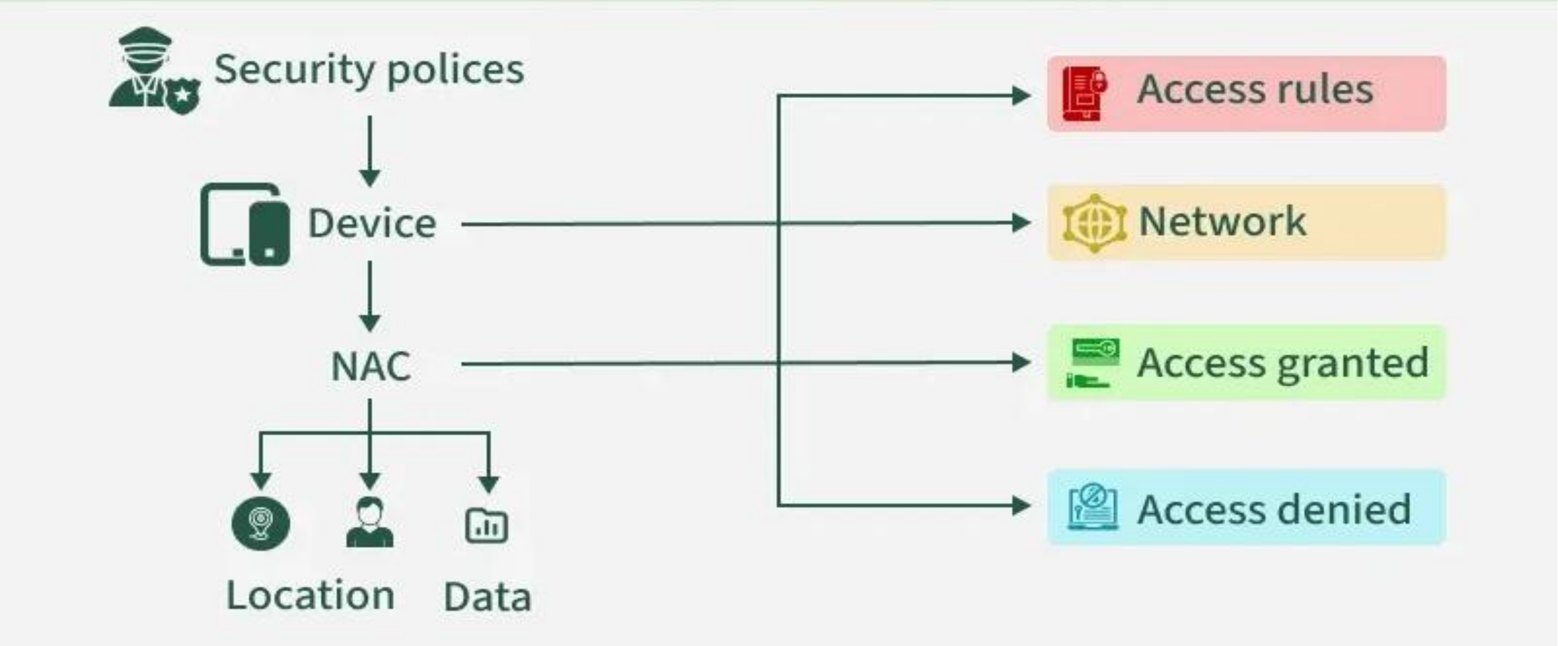


Network Access Control (NAC)

Think of it as the bouncer at the network's door

What Does NAC Actually Do?

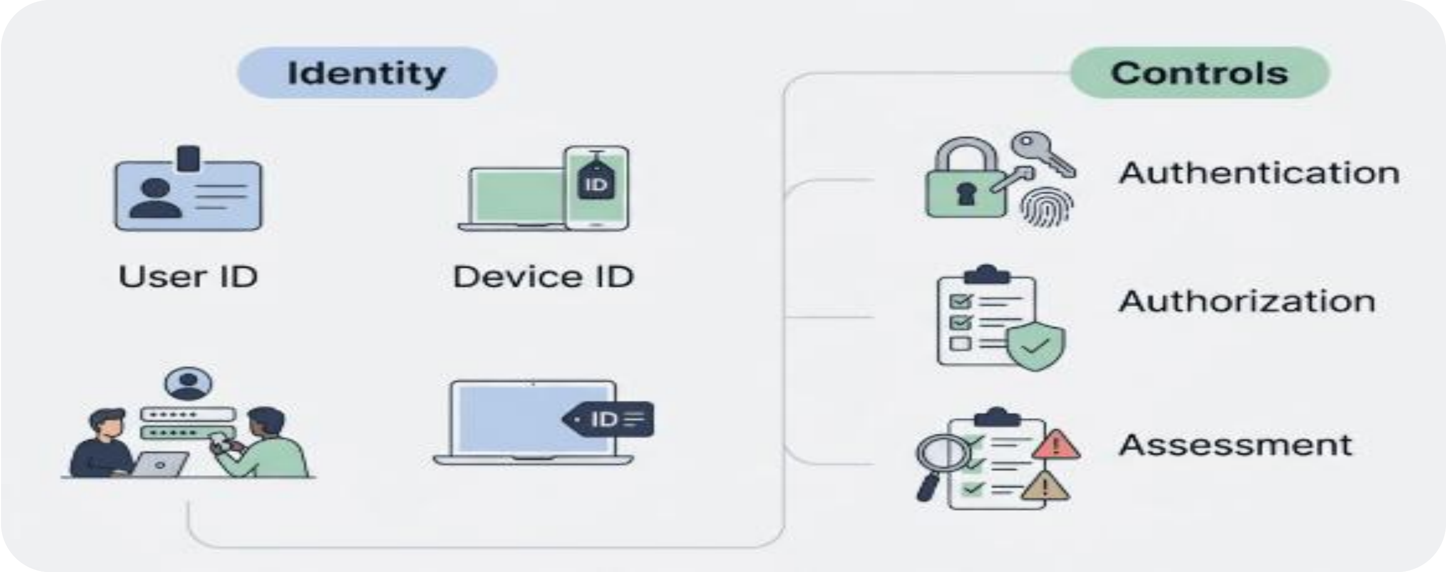
- Checks every user and device before they're allowed onto the network
- Without it, there's no realistic way to manually screen thousands of devices
- It's automatic — the rules get enforced without someone checking each device by hand



Why NAC Matters

Keeping every device honest before it connects

- Recognizes and profiles devices trying to connect
- Gives guests safe, limited access — usually through a registration portal
- Checks whether a device is compliant (updated, safe) before letting it on
- Can isolate or block a device that fails the check
- A key piece of zero trust — it enforces "never trust, always verify" for devices



Quick Quiz Review

Let's see if it stuck

Q1. **Which model looks at attributes like the user, the resource, and even the time of day?**

Answer: ABAC (Attribute-Based Access Control) — it checks several attributes at once, not just one factor.

Q2. **Which model bases access purely on someone's job role in the organization?**

Answer: RBAC (Role-Based Access Control) — also called non-discretionary access control.

Q3. **Which model is the strictest, typically used in military or mission-critical systems?**

Answer: MAC (Mandatory Access Control) — access is based on fixed security-level clearances.

Key Takeaways

What I really want you to remember

- Zero trust means verifying every request — never assume trust just because someone got in before
- There are several access control models (DAC, MAC, RBAC, ABAC, and more), each suited to different needs
- Least privilege limits damage; privilege escalation is the attack that tries to undo it
- NAC keeps devices honest by checking them before they ever touch the network
- Together, these ideas are what make real-world access control actually work